



Where We Live Gets Under the Skin: NaNDA and Immune Aging

GSA Annual Scientific Meeting Online Workshop Series 2026

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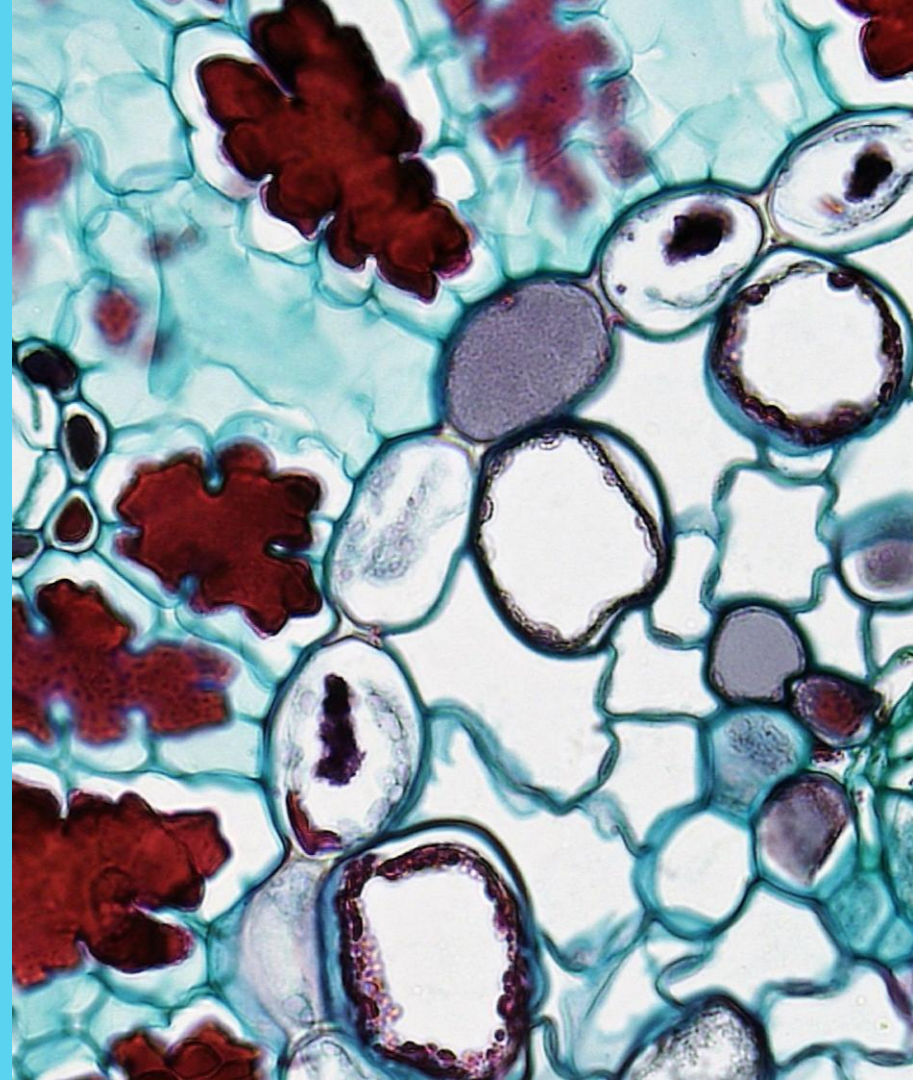


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Where We Live Gets Under the Skin:

NaNDA and Immune Aging

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**Can the neighborhood you
live in change the way
your immune system
ages?**

Why look at the immune system?

- Our immune system ages
- An “older” immune system is less able to fight off new infections and is associated with many chronic conditions
- Chronic stress and repeated infections may speed that aging up
- Neighborhoods may be a key source of stress and/or infectious diseases

Three signs of immune aging

1

CD4+ “helper” T cells

The balance of worn, experienced cells to fresh, ready-for-anything (naïve) cells. Measured as a ratio of older: younger cells.

2

CD8+ “killer” T cells

The same balance: more experienced cells relative to naïve cells means an older profile. Measured as a ratio of older: younger cells.

3

CMV antibodies

CMV is a common virus that stays in the body for life. Higher antibody levels suggest the immune system is working harder to keep it in check.

For all three, higher values point to an “older” immune profile.

The data: Health and Retirement Study



- Nationally representative study of Americans over 50
- Running since 1992, with follow-up every two years
- More than 20,000 participants
- In 2016, blood samples measured immune function
- We also knew where people lived in 2016

9,000+

older adults in each analysis with both immune data and a neighborhood we could link to NaNDA

How the NaNDA link works



The census tract ID is the bridge: one shared column links a health study to NaNDA.

What we pulled from NaNDA

Neighborhood disadvantage

- % living below the poverty line
- % unemployed
- % receiving public assistance or food stamps
- % female-headed households with children

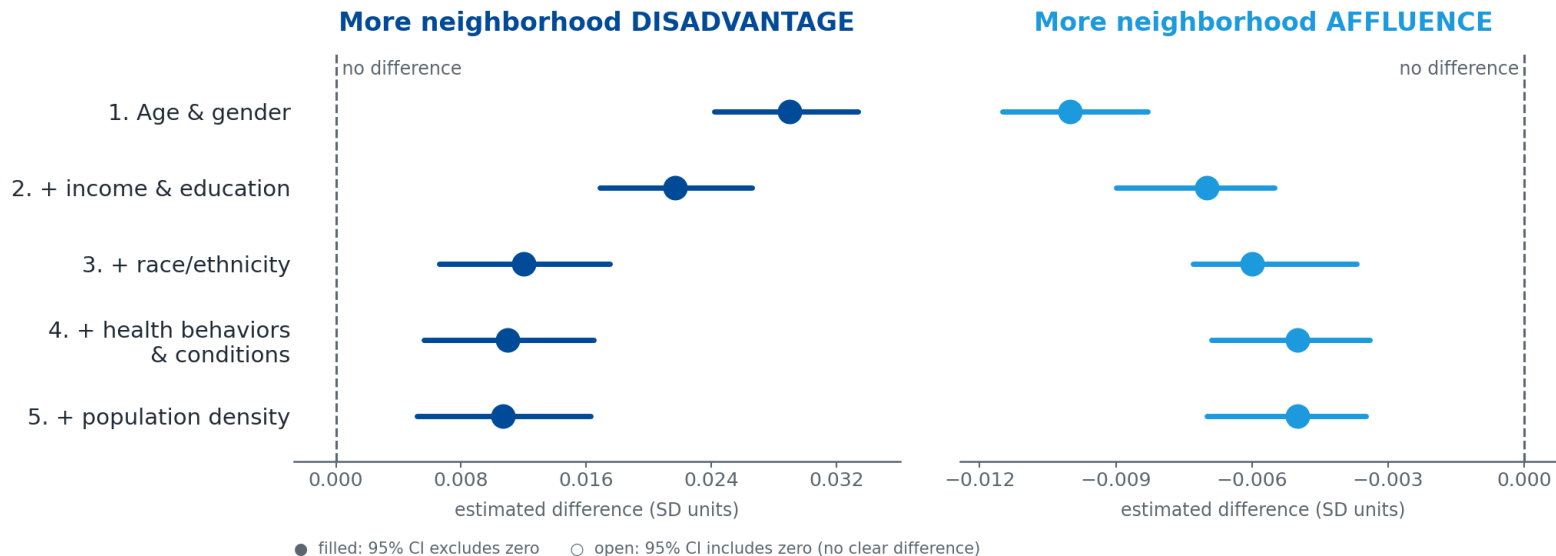
Neighborhood affluence

- % households earning over \$75K
- % in professional or managerial jobs
- % with a bachelor's degree or higher

Also: population density

How crowded the neighborhood is

What we found: CMV antibodies



More disadvantage → higher CMV antibodies. More affluence → lower.

The link shrinks once we account for people's own circumstances, but it never goes away.

Where the signal showed up

More disadvantage

More affluence

CD4+ T cells: experienced vs. naïve

▲ Higher: older profile*

No clear difference

CD8+ T cells: experienced vs. naïve

No clear difference

No clear difference

CMV antibodies

▲ Higher: older profile

▼ Lower: younger profile

Both CMV links held after accounting for personal income, education, race/ethnicity, health, and population density.

*Held after accounting for income and education; weakened once race/ethnicity was added.

**Where we live becomes
biologically embedded. It
shows up in our immune
systems.**

Why NaNDA made this possible

- Ready-made neighborhood measures for the whole U.S.
- Links to any study with a geographic ID, like a census tract
- Consistent definitions, so results can be compared across studies
- Lets health researchers examine whether where you live shows up in how you age

Thank you! Questions?

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Backup slides

- Detailed results: CD4+ T cells
- Detailed results: CD8+ T cells
- Who was in the study

Choose your breakout

Back together at 1:45

BEGINNER

Choosing Your First NaNDA Dataset

Pick this if

You are new to neighborhood data, or want help finding the right dataset.

You'll leave with

A NaNDA dataset matched to your geography and years.

INTERMEDIATE

Linking NaNDA with Your Data

Pick this if

Your data have addresses, ZIP codes, or census tracts to merge with NaNDA.

You'll leave with

A merge you ran on practice data, plus code you can reuse.

How to join

1

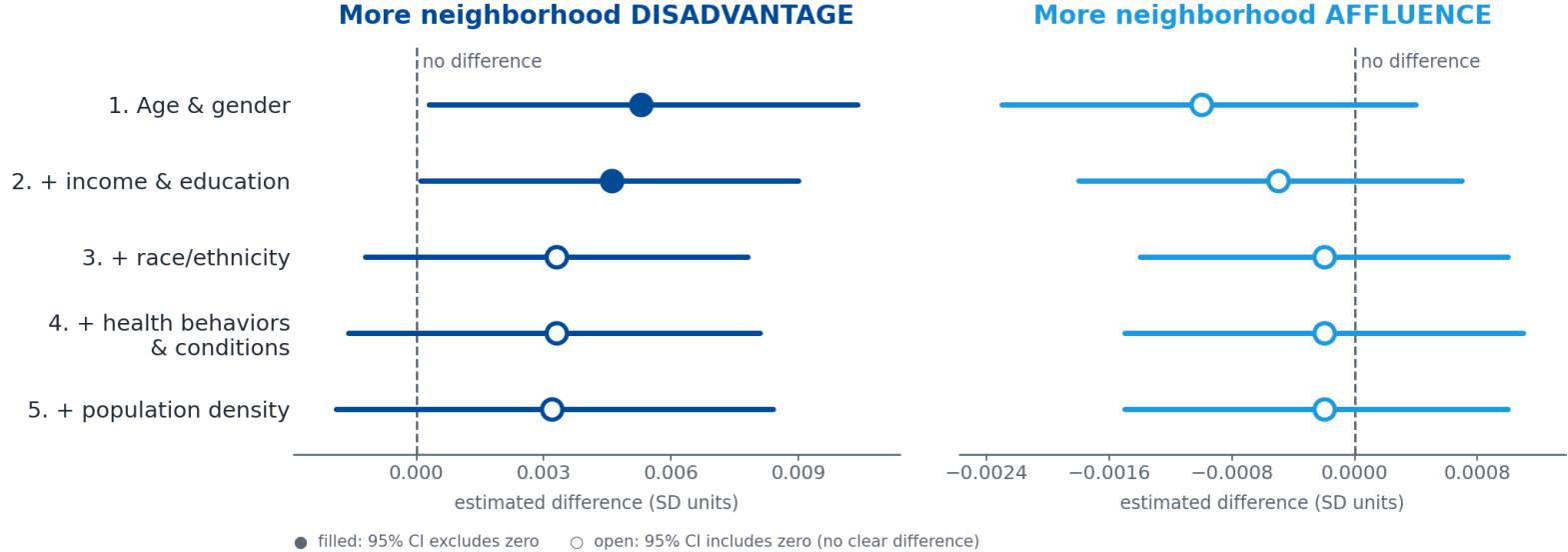
Click **Breakout Rooms** in your Zoom toolbar. It may be under **More**.

2

Choose **Beginner** or **Intermediate**, then click **Join**.

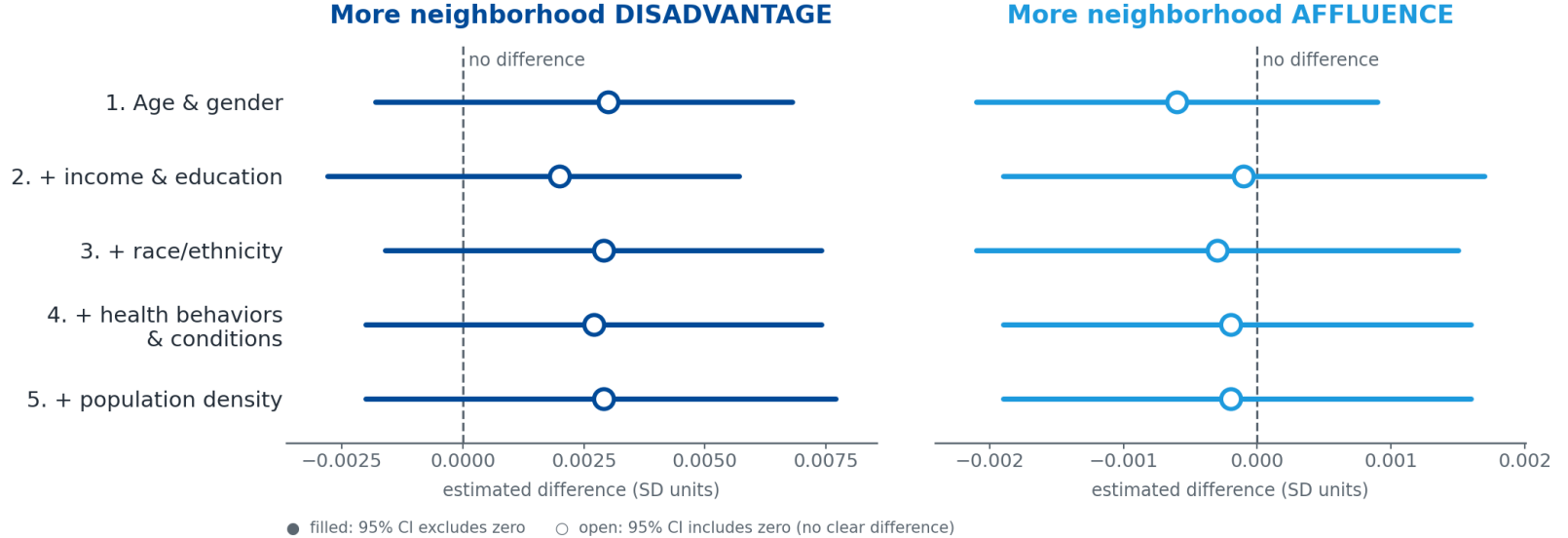
Not sure? Start with Beginner. You can switch rooms any time.

Backup: CD4+ T cells (EMRA:naïve ratio)



Linear regression estimates with 95% confidence intervals; outcomes standardized (SD units). Model 1: age + gender; Model 2: + income & education; Model 3: + race/ethnicity; Model 4: + health behaviors + chronic conditions; Model 5: + neighborhood population density.

Backup: CD8+ T cells (EMRA:naïve ratio)



Linear regression estimates with 95% confidence intervals; outcomes standardized (SD units). Model 1: age + gender; Model 2: + income & education; Model 3: + race/ethnicity; Model 4: + health behaviors + chronic conditions; Model 5: + neighborhood population density.

Backup: Who was in the study (N = 9,933)

Demographics

- Median age 67; 59% female
- 64% NH White, 18% NH Black, 15% Hispanic, 3% other
- Education: 17% less than HS, 51% HS grad, 7% some college, 24% college grad

Neighborhood, median (IQR)

- Disadvantage: 7.81 (7.5)
- Affluence: 28.37 (20.89)

Immune measures, median (IQR)

- CD8+:CD4+ ratio: 0.33 (0.31)
- CD4+ EMRA:naïve: 0.04 (0.12)
- CD8+ EMRA:naïve: 2.31 (4.88)
- CMV: 72% seropositive; IgG 214.2 U/mL (578.25)